



Target detection in hyperspectral imaging combining replacement and additive models

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ARTICLE INFO

Article history:

Received 24 March 2021

Revised 11 June 2021

Accepted 14 June 2021

Available online 16 June 2021

Keywords:

Hyperspectral imaging

Replacement model

GLRT

Target detection

ABSTRACT

In hyperspectral imaging the replacement model where a target, if present, partly replaces the disturbance is often advocated. In this paper, we consider a somehow more realistic model where only the low-rank background is substituted for the target while a residual noise, which belongs to the orthogonal complement, is unaffected by the presence/absence of the target. A two-step generalized likelihood ratio test is formulated for such a model. Furthermore we show that the log likelihood can be well approximated by a weighted combination of the log likelihoods of the FTMF and the AMF, and that the dimension of the background subspace is the tuning parameter which enables to balance between these two well-known detectors. A comparison with standard techniques on real hyperspectral data reveals a good performance of the new detectors.

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1. Introduction

Compared to standard imaging systems, an hyperspectral sensor can retrieve a precise spectral signature, composed of hundreds narrowbands, in each pixel. This feature allows to recognize the different components within the area of interest. Thereby, HyperSpectral Imaging (HSI) has encountered a large field of applications, ranging from remote sensing to medicine [1–8]. One of the main application consists in detection of a target whose signature is known [9–13]. Nevertheless, the spectral signatures measured at the remote sensor (satellite or airplane) cannot be directly compared to any target signatures database [14]. Indeed, many effects are likely to modify the reflectance of the area of interest. Among them, we can list the non-uniform sun illumination, the atmospheric absorption and scattering, but also the sensor transfer function itself. Thereby, a radiometric correction has usually to be conducted before processing any HSI.

To this end, two kinds of compensation can be conducted. The first one consists in inverting radiometric models, such as the popular MODTRAN [15]. The second one tends to identify specific ground targets with known reflectance, in order to estimate the coefficients of the supposed linear relationship between the radiance measured at the sensor level and the reflectance coming from the ground. The so-called Empirical Line Method (ELM) [15,16] is the best example of such a technique.

Hence, after radiometric compensation, the data represents the reflectance at ground level. As a consequence, if a sub-pixel target with known signature $\mathbf{t}$ is present in a Pixel Under Test (PUT) $\mathbf{y}$, it will mask a part of the background reflectance $\mathbf{b}$, and the so-called replacement model is usually advocated, namely $\mathbf{y} = \alpha\mathbf{t} + (1 - \alpha)\mathbf{b}$, where α represents the replacement fraction. If this so-called fill factor α is small, the simplified and widely used additive model can be used, namely $\mathbf{y} = \alpha\mathbf{t} + \mathbf{b}$.

Nevertheless, using such a popular replacement model raises different issues. Indeed, this model assumes a perfect radiometric compensation. Thereby, the PUT $\mathbf{y}$ is a simple linear mixture between of the target and the background, so that when the target is a full pixel target ($\alpha = 1$), there is no more stochastic part in the PUT, which is not totally realistic for real life measurements. Moreover, as the background is usually composed of a small number of components, so-called endmembers, the background belongs to a low-rank subspace. Then, as the amplitudes of each endmember, so-called abundances, are usually assumed to be stochastic, the background covariance matrix has also a low-rank structure, which is another problem, as most popular detectors exploit the inverse of such a matrix, that is singular in such a case.

In this paper, we propose to consider a more realistic model, which alleviates the above-mentioned problems by adding a residual part $\mathbf{n}$ to the replacement model, namely $\mathbf{y} = \alpha\mathbf{t} + (1 - \alpha)\mathbf{b} + \mathbf{n}$. The existence of this residual additive noise as been raised in some papers showing that at best, one can reach a 5% reflectance accuracy, using well-tuned radiometric compensation methods [17]. Indeed, there always remains a reflectance error

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due to non-modelled effects such as spatial illumination changes or unknown target's attitude. Then, it seems relevant to model this residual error using a zero-mean Gaussian distributed vector, namely $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_n)$. This way, the model proposed in this paper becomes a mixed replacement and additive model, and we will see here-after how it can generalize them.

Based on this more general and realistic model, we will derive, in this paper, the generalized likelihood ratio test (GLRT) for the problem at hand. To this end, we first state the hypotheses associated with this problem, in Section 2, and the exact GLRT formulation is derived in Section 3. Then, in order to simplify this expression, we propose to rewrite the problem at hand in a more relevant way in Section 4. This new formulation allows to derive a closed-form and asymptotically efficient expression of the fill factor. This simpler estimation conducts to a closed-form and more explainable expression of the test, that can be related to the GLRT of the additive and the replacement models, namely the Adaptive Matched Filter (AMF) [18] and the Finite Target Matched Filter (FTMF) [19]. To finish with, we compare the new detectors to state of the art algorithms, on a real HSI benchmarking, in Section 5. Concluding remarks end this paper in Section 6.

2. Detection problem statement

As stated in the introduction, the target detection problem can be written as a two hypotheses test, based on the following more realistic mixed additive and replacement model:

$$\begin{cases} H_0 & \mathbf{y} = \mathbf{b} + \mathbf{n} \\ H_1 & \mathbf{y} = \alpha \mathbf{t} + (1 - \alpha) \mathbf{b} + \mathbf{n} \end{cases} \quad (1)$$

where both the background $\mathbf{b}$ and the residual noise $\mathbf{n}$ are assumed to be independent Gaussian distributed vectors, namely $\mathbf{b} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}_b)$ and $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_n)$. As usually assumed in this context, the target signature $\mathbf{t}$ is supposed to be known, and the background and noise can be learnt from secondary pixels assumed to be distributed as $\mathbf{y}$ under H_0 , namely $\mathbf{z}_k \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}_b + \mathbf{C}_n)$ for $k = 0, \dots, (K - 1)$.

As stated in the introduction, it is usually assumed that the background is composed of a small number of endmembers, so that $\mathbf{C}_b$ is a low-rank matrix that can be written as $\mathbf{C}_b = \mathbf{U}_s \boldsymbol{\Lambda}_s \mathbf{U}_s^T$, where the columns of $\mathbf{U}_s$ are the r eigenvectors and $\boldsymbol{\Lambda}_s$ is the diagonal matrix composed of the r eigenvalues of $\mathbf{C}_b$. On the other hand, $\mathbf{C}_n$ is assumed to be full-rank and can be written as $\mathbf{C}_n = \mathbf{U}_n \boldsymbol{\Lambda}_n^s \mathbf{U}_n^T + \mathbf{U}_n \boldsymbol{\Lambda}_n \mathbf{U}_n^T$. As $\mathbf{n}$ is only a residual error, we will assume hereafter that $\boldsymbol{\Lambda}_n^s$ is negligible compared to $\boldsymbol{\Lambda}_n$. This assumption amounts to neglect the energy of the projection of $\mathbf{n}$ onto the background subspace compared to the background energy, which is a far less restrictive assumption than in the conventional replacement model where $\mathbf{n}$ is simply null.

In this paper we focus on a so-called two-step approach. That is, we assume that both the background and noise parameters are known from the secondary pixels $\mathbf{z}_k$. Indeed, as $\mathbf{z}_k \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C}_b + \mathbf{C}_n)$, we can easily estimate both the mean and the covariance matrix, in a Maximum Likelihood (ML) sense as $\hat{\boldsymbol{\mu}} = \frac{1}{K} \sum_{k=0}^{K-1} \mathbf{z}_k$ and $\hat{\mathbf{C}}_z = \frac{1}{K} \sum_{k=0}^{K-1} (\mathbf{z}_k - \hat{\boldsymbol{\mu}})(\mathbf{z}_k - \hat{\boldsymbol{\mu}})^T$. Then, using eigen-decomposition, we can estimate $\mathbf{U}_s$, $\mathbf{U}_n$, $(\boldsymbol{\Lambda}_s + \boldsymbol{\Lambda}_n^s)$ and $\boldsymbol{\Lambda}_n$ for a given r . A discussion about how to choose r as well as its influence on the detector structure will be developed in Section 4.

3. GLRT Derivation

Assuming that the residual noise energy is negligible in the background subspace ($\boldsymbol{\Lambda}_n^s \ll \boldsymbol{\Lambda}_s$), as explained in the previous section, the data under H_1 are distributed as $\mathbf{y} \sim$

$\mathcal{N}(\alpha \mathbf{t} + (1 - \alpha) \boldsymbol{\mu}, (1 - \alpha)^2 \mathbf{U}_s \boldsymbol{\Lambda}_s \mathbf{U}_s^T + \mathbf{U}_n \boldsymbol{\Lambda}_n \mathbf{U}_n^T)$. Then, the log likelihood writes

$$\begin{aligned} L(\alpha; \mathbf{y}) \simeq & -\frac{1}{2} [N \log 2\pi + \log((1 - \alpha)^{2r} |\boldsymbol{\Lambda}_s|) + \log(|\boldsymbol{\Lambda}_n|) \\ & + (\mathbf{y} - \alpha \mathbf{t} - (1 - \alpha) \boldsymbol{\mu})^T (\mathbf{U}_s (1 - \alpha)^2 \boldsymbol{\Lambda}_s \mathbf{U}_s^T + \mathbf{U}_n \boldsymbol{\Lambda}_n \mathbf{U}_n^T)^{-1} \\ & \times (\mathbf{y} - \alpha \mathbf{t} - (1 - \alpha) \boldsymbol{\mu})] \end{aligned} \quad (2)$$

or, more simply

$$L(\alpha; \mathbf{y}) \simeq -r \log(1 - \alpha) - \frac{\|\mathbf{y}_s - \alpha \mathbf{t}_s\|^2}{2(1 - \alpha)^2} - \frac{1}{2} \|\mathbf{y}_n - \alpha \mathbf{t}_n\|^2 + \text{const.} \quad (3)$$

where $\|\cdot\|$ stands for the Euclidean norm of the vector, $\mathbf{y}_s = \boldsymbol{\Lambda}_s^{-\frac{1}{2}} \mathbf{U}_s^T (\mathbf{y} - \boldsymbol{\mu})$, $\mathbf{t}_s = \boldsymbol{\Lambda}_s^{-\frac{1}{2}} \mathbf{U}_s^T (\mathbf{t} - \boldsymbol{\mu})$, $\mathbf{y}_n = \boldsymbol{\Lambda}_n^{-\frac{1}{2}} \mathbf{U}_n^T (\mathbf{y} - \boldsymbol{\mu})$ and $\mathbf{t}_n = \boldsymbol{\Lambda}_n^{-\frac{1}{2}} \mathbf{U}_n^T (\mathbf{t} - \boldsymbol{\mu})$ are the centred and whitened versions of the projections of both the data and the target onto the noise and background subspaces.

In order to derive the GLRT, we need to estimate the fill factor α . To this end, we have to maximize this log likelihood with respect to α . Differentiating (3) with respect to α and setting the result to zero leads to the following 4th order equation:

$$\begin{aligned} & (\mathbf{t}_n^T \mathbf{t}_n) (1 - \alpha)^4 + (\mathbf{t}_n^T (\mathbf{y}_n - \mathbf{t}_n)) (1 - \alpha)^3 + r (1 - \alpha)^2 \\ & - (\mathbf{t}_s^T (\mathbf{y}_s - \mathbf{t}_s)) (1 - \alpha) - \|\mathbf{y}_s - \mathbf{t}_s\|^2 = 0 \end{aligned} \quad (4)$$

Then, $\hat{\alpha}_{ML}$, the ML estimation of α can be obtained by finding the roots of such an equation, but no simple closed-form formulation exists. It can be noticed that when $r = 0$, $\mathbf{y}_s = \mathbf{t}_s = \mathbf{0}$, and Eq. (4) reduces to the first order equation associated with the standard additive model. On the other hand, when $r = N$, $\mathbf{y}_n = \mathbf{t}_n = \mathbf{0}$, and Eq. (4) reduces to that of the FTMF derivation [20], which comforts the validity of Eq. (4).

Then, the GLRT for the problem at hand writes

$$GLRT = L(\hat{\alpha}_{ML}; \mathbf{y}) - L(0; \mathbf{y}) \quad (5)$$

Although optimal, this non closed-form expression is also difficult to analyse in a comprehensive way. In the next section, we will give an equivalent closed-form expression (eq. (15)) for this GLRT, that is more easily related to both AMF and FTMF.

4. Approximate GLRT and insights

Inspecting Eq. (3), we can see that $L(\alpha; \mathbf{y}) = L_r(\alpha; \mathbf{y}_s) + L_a(\alpha; \mathbf{y}_n) + \text{const.}$ where

$$L_r(\alpha; \mathbf{y}_s) = -r \log(1 - \alpha) - \frac{\|\mathbf{y}_s - \alpha \mathbf{t}_s\|^2}{2(1 - \alpha)^2} \quad (6a)$$

$$L_a(\alpha; \mathbf{y}_n) = -\frac{1}{2} \|\mathbf{y}_n - \alpha \mathbf{t}_n\|^2 \quad (6b)$$

correspond respectively to the log likelihood of a replacement and an additive model. Indeed, when projecting the data $\mathbf{y}$ onto both the background subspace $\mathbf{U}_s$ and the noise subspace $\mathbf{U}_n$, and after the whitening steps by $\boldsymbol{\Lambda}_s^{-\frac{1}{2}}$ and $\boldsymbol{\Lambda}_n^{-\frac{1}{2}}$, we have the following equivalent model for the data:

$$\mathbf{y}_s = \alpha \mathbf{t}_s + (1 - \alpha) \mathbf{b}_s \quad (7a)$$

$$\mathbf{y}_n = \alpha \mathbf{t}_n + \mathbf{n}_n \quad (7b)$$

where $\mathbf{b}_s = \boldsymbol{\Lambda}_s^{-\frac{1}{2}} \mathbf{U}_s^T (\mathbf{b} - \boldsymbol{\mu}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{n}_n = \boldsymbol{\Lambda}_n^{-\frac{1}{2}} \mathbf{U}_n^T \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and we have to decide whether $\alpha = 0$ or not.

This new writing of the model is then equivalent to the initial model, and the likelihood of observing both $\mathbf{y}_s$ and $\mathbf{y}_n$ remains

$L(\alpha; \mathbf{y}) = L_r(\alpha; \mathbf{y}_s) + L_a(\alpha; \mathbf{y}_n)$. But the benefits of such a subspace decomposition is to easily link the initial problem to more familiar ones, namely a replacement and an additive model.

Now, in order to simplify the estimation of α , we can exploit this natural splitting of the general model. To this end, we can refer to the EXtended Invariance Principle (EXIP) [21,22]. This procedure allows to replace a challenging maximization by two simpler steps, while preserving the efficiency of the ML estimation, at least asymptotically. This extension of the popular ML reparametrization to non-bijective functions has been widely used in many areas, sometimes without explicitly naming it [23–29]. Its principle is shortly described in the following sub-section.

4.1. EXIP principle

The main idea behind EXIP is a re-parametrization of the problem at hand whose goal is to simplify the minimization. Then, we can estimate the initial parameters by solving an optimally weighted least-squares minimization problem. More precisely, assume one wants to minimize a function $V(\boldsymbol{\theta})$ ($V(\boldsymbol{\theta}) = -L(\boldsymbol{\theta})$, in our case of interest) where some structure is imposed on $\boldsymbol{\theta}$. Often, relaxing the constraints by reparameterizing the problem in terms of a less structured vector $\boldsymbol{\eta} = f(\boldsymbol{\theta})$ leads to a simple solution, say $\hat{\boldsymbol{\eta}}$. Then, one can get an asymptotically equivalent solution on $\boldsymbol{\theta}$ from the intermediate estimate $\hat{\boldsymbol{\eta}}$ as:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} (\hat{\boldsymbol{\eta}} - f(\boldsymbol{\theta}))^T \mathbf{W} (\hat{\boldsymbol{\eta}} - f(\boldsymbol{\theta})) \quad (8)$$

where

$$\mathbf{W} = \mathbb{E} \left\{ \frac{\partial^2 V(\boldsymbol{\eta})}{\partial \boldsymbol{\eta} \partial \boldsymbol{\eta}^T} \right\} \quad (9)$$

The goal of EXIP is to replace a challenging maximization by two simpler ones, while preserving asymptotically the performance.

4.2. Fill factor estimation using EXIP

In our case of interest, referring to Eq. (7), we first relax the constraints between the two sub-models, defining $\boldsymbol{\eta} = [\alpha_r \ \alpha_a]^T$, where α_r is the fill factor for $\mathbf{y}_s$ and α_a is a different fill factor for $\mathbf{y}_n$. This way, we can separately estimate the fill factor for the replacement model on one hand, and for the additive model on the other hand, as they should be the same according to model (7). Then, we will refine the estimation of α thanks to the mean least square optimization (8). The fill factors for each model are shown to be [20]:

$$\hat{\alpha}_r = 1 - \frac{\mathbf{t}_s^T (\mathbf{y}_s - \mathbf{t}_s)}{2N} - \frac{\sqrt{(\mathbf{t}_s^T (\mathbf{y}_s - \mathbf{t}_s))^2 + 4N(\mathbf{y}_s - \mathbf{t}_s)^T (\mathbf{y}_s - \mathbf{t}_s)}}{2N} \quad (10)$$

$$\hat{\alpha}_a = \frac{\mathbf{t}_n^T \mathbf{y}_n}{\mathbf{t}_n^T \mathbf{t}_n}$$

where $\hat{\alpha}_r$ and $\hat{\alpha}_a$ are respectively the fill factors derived in the FTMF [19] and the AMF [18] detectors.

Now to integrate back the links between α_r and α_a , namely $\boldsymbol{\eta} = f(\alpha) = \alpha \mathbf{1}$, and get an asymptotically efficient estimation of α , we have to solve the following least squares problem, thanks to EXIP procedure:

$$\hat{\alpha}_{EXIP} = \arg \min_{\alpha} (\hat{\boldsymbol{\eta}} - \alpha \mathbf{1})^T \mathbf{W} (\hat{\boldsymbol{\eta}} - \alpha \mathbf{1}) \quad (11)$$

where

$$\mathbf{W} = -\mathbb{E} \left\{ \frac{\partial^2 L(\boldsymbol{\eta}; \mathbf{y})}{\partial \boldsymbol{\eta} \partial \boldsymbol{\eta}^T} \right\} = \begin{pmatrix} -\mathbb{E} \left\{ \frac{\partial^2 L_r(\alpha_r; \mathbf{y}_s)}{\partial \alpha_r^2} \right\} & 0 \\ 0 & -\mathbb{E} \left\{ \frac{\partial^2 L_a(\alpha_a; \mathbf{y}_n)}{\partial \alpha_a^2} \right\} \end{pmatrix}$$

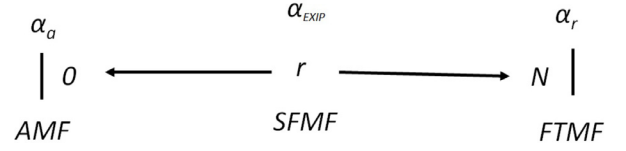


Fig. 1. SFMF understanding vs the background rank r .

and $\hat{\boldsymbol{\eta}} = [\hat{\alpha}_r \ \hat{\alpha}_a]^T$. It is straightforward to show that $\mathbf{W}$ writes

$$\mathbf{W} = \begin{pmatrix} w_r & 0 \\ 0 & w_a \end{pmatrix} \quad (12)$$

where $w_r = \frac{\mathbf{t}_s^T \mathbf{t}_s + 2r}{(1 - \hat{\alpha}_r)^2}$ and $w_a = \mathbf{t}_n^T \mathbf{t}_n$.

Thereby, the solution of Eq. (11) is direct, and gives

$$\hat{\alpha}_{EXIP} = \frac{w_r \hat{\alpha}_r + w_a \hat{\alpha}_a}{w_r + w_a} \quad (13)$$

This asymptotically efficient estimation of α depends on the rank of the background subspace r that can be estimated from the sample covariance matrix of the secondary pixels. Indeed, we assume that the eigenvalues of Λ_n are smaller than those of Λ_s , as Λ_n represents only a residual error. Then, different techniques can be used to estimate the two subspaces, such as the popular Akaike Information Criterion (AIC) [30], for instance. As presented in Section 5, we consider here a subspace splitting simply based on an energy criterion.

On the other hand, r can also be considered as a tuning parameter to choose between a full replacement model if $r = N$ and a full additive model if $r = 0$. Indeed, in the first case, the noise subspace is null and Eq. (7) resumes to a replacement model, and $w_a = 0$ so that $\hat{\alpha}_{EXIP} = \hat{\alpha}_r$. It is the opposite in the second case, where the simpler additive model prevails on all the space, and $\hat{\alpha}_{EXIP} = \hat{\alpha}_a$, as $w_r = 0$.

To finish with, as the criterion to be maximized is a log likelihood function, it can be noticed that w_r^{-1} and w_a^{-1} are also the Cramer-Rao Bounds (CRB) for the fill factors in a replacement and an additive case.

4.3. Subspace fitting matched filter

Given this simpler estimation for α , the GLRT now writes

$$\begin{aligned} SFMF &= L(\hat{\alpha}_{EXIP}; \mathbf{y}) - L(0; \mathbf{y}) \\ &= [L_r(\hat{\alpha}_{EXIP}; \mathbf{y}_s) - L_r(0; \mathbf{y}_s)] + [L_a(\hat{\alpha}_{EXIP}; \mathbf{y}_n) - L_a(0; \mathbf{y}_n)] \end{aligned} \quad (14)$$

where we recall that L_r and L_a are the Log likelihoods for the replacement and additive part, respectively, and that $\hat{\alpha}_{EXIP} \rightarrow \hat{\alpha}_r$ when $r \rightarrow N$, and $\hat{\alpha}_{EXIP} \rightarrow \hat{\alpha}_a$ when $r \rightarrow 0$, and that $L_r(\alpha; \mathbf{y})$ or $L_a(\alpha; \mathbf{y})$ are 0 for each of these two cases. Hence, the proposed test also tends toward FTMF or AMF depending on the choice on the covariance matrix rank r , namely $SFMF \rightarrow [L_r(\hat{\alpha}_r; \mathbf{y}) - L_r(0; \mathbf{y})] = FTMF$ when $r \rightarrow N$ and $SFMF \rightarrow [L_a(\hat{\alpha}_a; \mathbf{y}) - L_a(0; \mathbf{y})] = AMF$ when $r \rightarrow 0$. Then, r can be seen as a tuning parameter to favour replacement or additivity, as shown on Fig. 1. That is the reason why we choose to name this new detector as Subspace Fitting Matched Filter (SFMF).

Moreover, FTMF and AMF are known to have different benefits. FTMF is known to be more selective than AMF [31,32]. Then in a complicated scenario, where possible target-like background components may exist, using the FTMF will be better than AMF. On the contrary, AMF is more robust and can improve detection in a more simple scenario. The proposed SFMF should keep these two advantages, as it is the best association of those two popular detectors, with respect to the realistic model proposed in this paper.

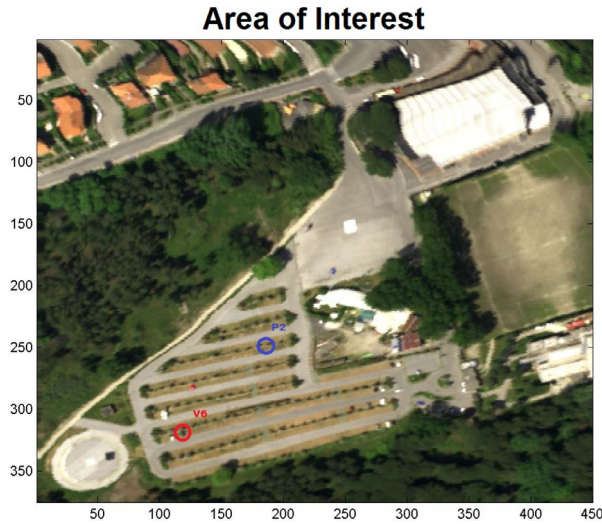


Fig. 2. RGB view of the Viareggio test scene.

Combining two detectors with different features is not a new approach and it has been studied in the radar field [33] but also in HSI field with the False Alarm Mitigation (FAM) techniques [34–38], for instance. Nevertheless, unlike in our case, these kinds of techniques use the same data as an input for the two detectors and one has to choose different thresholds and combine the two tests to take a final decision.

5. Performance evaluation

We now propose to compare the two kinds of detectors introduced in this paper to the state of the art ones, on a real-life benchmarking. To this end, we consider the airborne Viareggio 2013 trial [39] that took place in Viareggio (Italy), in May 2013, with an aircraft flying at 1200 meters. The image is composed of $[450 \times 375]$ pixels with 511 samples in the Visible Near InfraRed (VINR) band (400 – 1000nm).

Different kinds of vehicles as well as coloured panels served as known targets. For each of these targets, a spectral signature obtained from ground spectroradiometer measurements is available. As can be seen on Fig. 2, the scene is composed of parking lots, roads, buildings, sport fields and pine woods.

In addition, calibration targets are used to conduct an Empirical Line Method (ELM) [15,16] to convert the raw measurements into reflectance. Then a spectral binning [40] is performed to reduce the vector size dimension to $N = 64$.

In this benchmarking, we compare both SFMF and the GLRT based on the exact solution for α (Eq. (5)) with AMF [18], FTMF [19], obviously, but also with Kelly [41–43], ACE [44–46], the modified FTMF from [47] and SPADE [47]. We use the so-called false alarm scores to compare these algorithms, namely the number of pixels having their detector's output strictly higher than the one for the target pixel. This number can be seen as a false alarm number with an optimal thresholding.

We focus on two kinds of targets, namely a panel (P_2) and a vehicle (V_6), as can be seen on Fig. 2. Those two targets are a few meters length and the spatial resolution of the image is about 0.6 meters, so that we chose a guard window size of 9×9 pixels, in order to avoid the presence of target signature in the covariance matrix estimation window. At last, the rank r is estimated from the secondary pixels covariance matrix by thresholding the eigenvalues. Indeed, the residual noise energy being quite small, after the radiometric compensation, we chose the background so that

Background Subspace Estimated Rank (N=64)

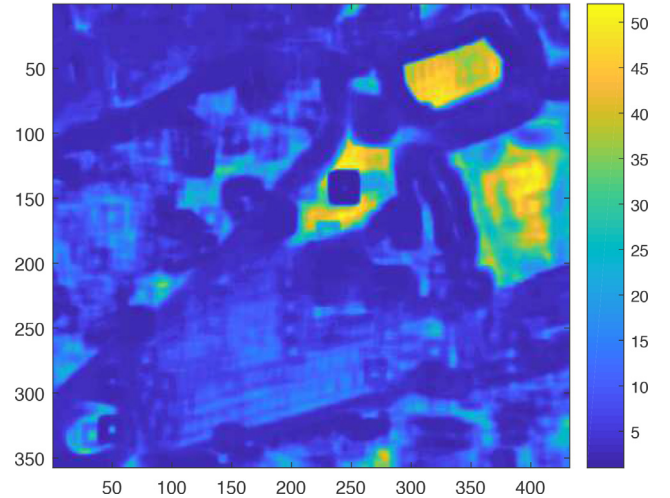


Fig. 3. Covariance matrix background subspace rank estimation on D2F12H2 image (19×19 window).

its energy is larger than 0.99 of the total energy. Fig. 3 gives an example of the estimated rank on the whole image for a 19×19 window. As can be seen, this rank varies between 1 and 52 with respect to the homogeneity of the area of interest in the image. However, the mean rank value remains low ($\bar{r} = 8.2$) with respect to the covariance matrix dimension $N = 64$.

Fig. 4 represents an example of some detectors outputs in the surroundings of P_2 target, which is located in the center of the maps. As expected, we can see that AMF presents a quite strong peak at the target location, but also possible false alarms peaks corresponding to target-like pixels. This last point is clearly visible on Kelly's output. FTMF lowers these target-like possible false alarms, but the target's peak is also reduced. The two GLRT proposed in this paper offer a very good compromise between detection performance and selectivity (false alarm mitigation), as they both have a high peak at the target location, while maintaining the level of the target-like pixel quite low.

This result can also be observed in Fig. 5 where the fill factors estimated by the different methods are represented for the same local area of interest. We can see that AMF finds $\alpha \approx 0.65$ at the target position, which is lower than the real value of α , as we can expect a full pixel target in the central pixel. This under-estimation is certainly linked to target signature variability and incomplete radiometric compensation, as mentioned in this paper. AMF also estimates strong target-like items with α up to 0.4 in non-target pixels. On the other side, these peaks are reduced with FTMF, but the target's peak is also reduced to $\alpha = 0.42$. Both the ML and the EXIP estimation of α , using the model proposed in this paper, are a mix between these two extremes, and increase the gap between the target and the target-like pixels. This fill factor estimation being the central point for the test performance, we can expect a good behaviour of the two associated detectors, as can be seen in the following tables.

Tables 1 and 2 present the false alarm scores for both the P_2 and the V_6 targets and different window sizes. First of all, we can observe a substantial gap in performance between the two targets, V_6 being more difficult to detect. Moreover, we can also observe a difference in the ranking of the methods between the target type. Indeed, ACE is the best algorithm on the V_6 target, whereas it is worse on the P_2 target. Hence, as it is already known, the performance of each method is closely related to the kinds of image and target. Nevertheless, the two proposed algorithms seems to perform well in all the cases, as they always rank among the best

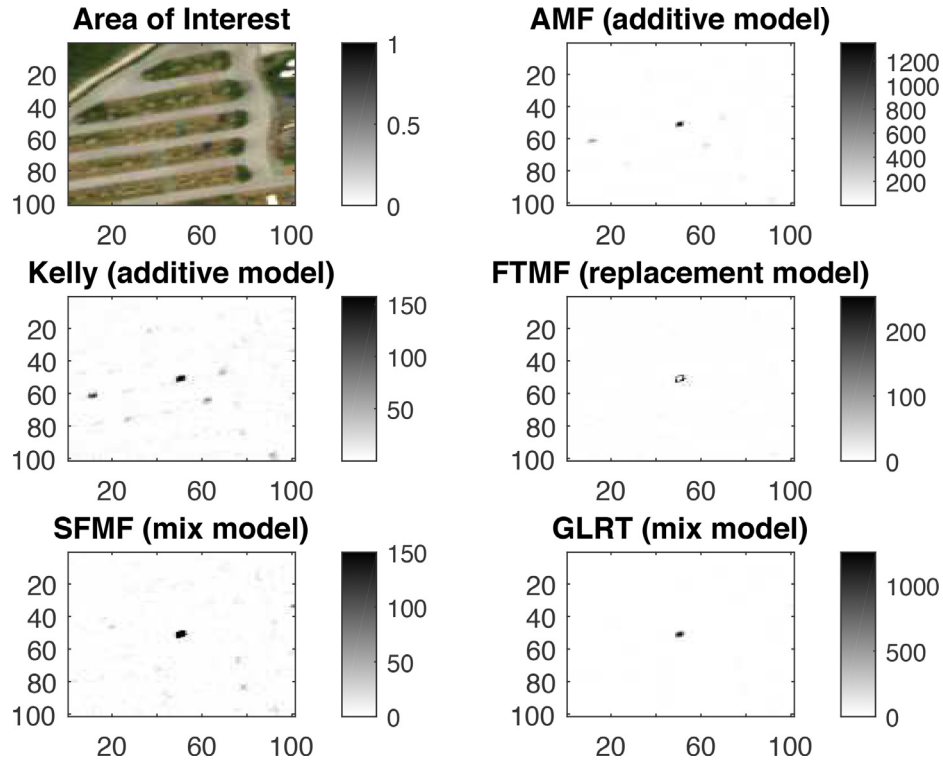


Fig. 4. detector's output near the target position.

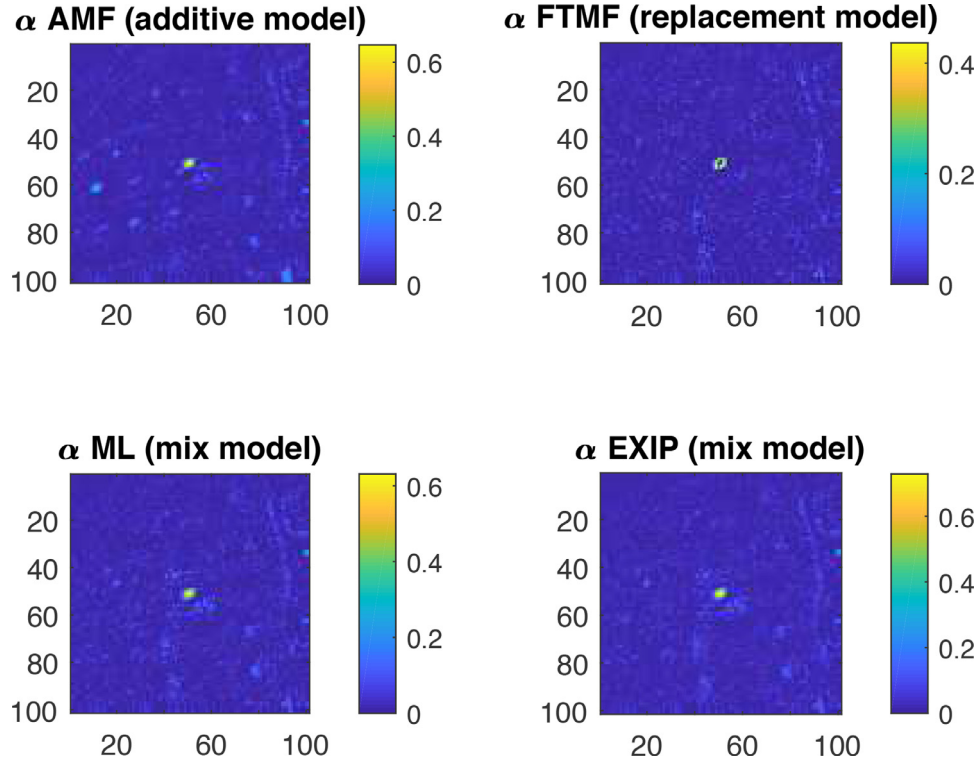


Fig. 5. fill factor estimations output near the target position.

Table 1

False Alarms score for P_2 target in the D1F12H2 Viareggio open data image .

Window Size	SFMF	Eq. (5)	AMF	FTMF	Kelly	ACE	Modif. FTMF	SPADE
17×17	1	1	3	7	3	7	3	3
19×19	0	0	2	5	2	4	2	2
21×21	0	0	0	4	0	2	0	0

Table 2False Alarms score for V_6 target in the D2F12H2 Viareggio open data image .

Window Size	SFMF	Eq. (5)	AMF	FTMF	Kelly	ACE	Modif. FTMF	SPADE
17×17	4	2	36	241	4	0	37	14
19×19	4	2	33	123	10	0	40	21
21×21	4	1	29	103	10	0	35	18

ones. The exact GLRT for the mixed model proposed in this paper (Eq. (5)) performs slightly better than the SFMF, but the price to be paid is a to solve a 4th order polynomial equation, whereas SFMF is closed-form.

6. Conclusions

In this paper, we considered target detection in an hyperspectral imaging context. We introduced a more realistic model than the conventional replacement model, and derived the corresponding GLRT. This model is shown to be approximately a mixed replacement and additive model, so that we proposed a simplified version of the GLRT that can be easily linked to both the FTMF and the AMF, which hold respectively for the replacement and for the additive model. As a consequence, this proposed detector, namely SFMF inherits the benefits of these two detectors, namely a good selectivity while preserving detection performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

François Vincent: Conceptualization, Methodology, Software, Writing - original draft. **Olivier Besson:** Methodology, Validation, Formal analysis, Writing - review & editing.

References

- [1] D.G. Manolakis, R.B. Lockwood, T.W. Cooley, *Hyperspectral Imaging Remote Sensing*, Cambridge University Press, 2016.
- [2] E. Keith, H. Dan, O. William, J. Shridhar, B. Eustace, L. Dereniak, *Hyperspectral imaging for astronomy and space surveillance*, in: Proc. SPIE, 5159, 2004.
- [3] S. Michel, P. Gamet, M.-J. Lefevre-Fonollosa, *Hypxim a hyperspectral satellite defined for science, security and defence users*, Proceedings 3rd Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (WHISPERS), 2011.
- [4] H. Kwon, N. Nasrabadi, Kernel rx-algorithm: a nonlinear anomaly detector for hyperspectral imagery, *IEEE Trans. Geosci. Remote Sens.* 43 (2) (2005) 388–397.
- [5] E.M. Winter, M. Miller, C. Simi, A. Hill, T. Williams, D. Hampton, M. Wood, J. Zadnick, M. Sviland, Mine detection experiments using hyperspectral sensors, *SPIE Int. Soc. Opt. Eng.*, 2004, Orlando, FL, United States.
- [6] C.C. Funk, J. Theiler, D.A. Roberts, C.C. Borel, Clustering to improve matched filter detection of weak gas plumes in hyperspectral thermal imagery, *IEEE Trans. Geosci. Remote Sens.* 39 (7) (2001) 1410–1420.
- [7] D.-W. Sun, *Hyperspectral Imaging for Food Quality Analysis and Control*, Elsevier, 2010.
- [8] R. Koprowski, *Processing of Hyperspectral Medical Images, Applications in Dermatology Using Matlab*, Springer International Publishing, 2017.
- [9] D. Manolakis, G. Shaw, Detection algorithms for hyperspectral imaging applications, *IEEE Signal Process. Mag.* 19 (1) (2002) 29–43.
- [10] J. Settle, On constrained energy minimization and the partial unmixing of multispectral images, *IEEE Trans. Geosci. Remote Sens.* 40 (3) (2002) 718–721.
- [11] C. Chang, Orthogonal subspace projection (osp) revisited: a comprehensive study and analysis, *IEEE Trans. Geosci. Remote Sens.* 43 (3) (2005) 502–518.
- [12] Y. Dong, T. Liang, Y. Zhang, B. Du, Spectral-spatial weighted kernel manifold embedded distribution alignment for remote sensing image classification, *IEEE Trans. Cybern.* 51 (6) (2021) 3185–3197, doi:10.1109/TCYB.2020.3004263.
- [13] L. Zhang, L. Zhang, D. Tao, X. Huang, Sparse transfer manifold embedding for hyperspectral target detection, *IEEE Trans. Geosci. Remote Sens.* 52 (2) (2014) 1030–1043, doi:10.1109/TGRS.2013.2246837.
- [14] Ecotress spectral library - version 1.0, 2017.
- [15] G. Ferrier, Evaluation of apparent surface reflectance estimation methodologies, *Int. J. Remote Sens.* 16 (1995) 2291–2297.
- [16] G.M. Smith, E.J. Milton, The use of the empirical line method to calibrate remotely sensed data to reflectance, *Int. J. Remote Sens.* 20 (1999) 2653–2662.
- [17] L. Markelin, E. Honkavaara, T. Takala, D. Schlöpfer, J. Suomalainen, P. Pellikka, A novel approach for the radiometric correction of airborne hyperspectral image data, 2012.
- [18] F.C. Robey, D.R. Fuhrmann, E.J. Kelly, R. Nitzberg, A CFAR adaptive matched filter detector, *IEEE Trans. Aerospace Electron. Syst.* 28 (1) (1992) 208–216.
- [19] A. Schaum, A. Stocker, Spectrally-selective target detection, in: Proceedings of ISSSR, 12, 1997, pp. 2015–2018.
- [20] F. Vincent, O. Besson, One-step generalized likelihood ratio test for subpixel target detection in hyperspectral imaging, *IEEE Trans. Geosci. Remote Sens.* 58 (6) (2020) 4479–4489.
- [21] P. Stoica, T. Soderström, On reparametrization of loss functions used in estimation and the invariance principle, *Signal Process.* 17 (4) (1989) 383–387.
- [22] A.L. Swindlehurst, P. Stoica, Maximum likelihood methods in radar array signal processing, *Proc. IEEE* 86 (2) (1998) 421–441, doi:10.1109/5.659495.
- [23] A. Amar, A.J. Weiss, Localization of narrowband radio emitters based on doppler frequency shifts 56 (11) (2008) 5500–5508.
- [24] A.J. Weiss, Direct position determination of narrowband radio frequency transmitters 11 (5) (2004) 513–516.
- [25] A.J. Weiss, A. Amar, Direct position determination of multiple radio signals, *EURASIP J. Appl. Signal Process.* (2005) 37–49.
- [26] F. Vincent, J. Vila-Valls, O. Besson, D. Medina, E. Chaumette, Doppler-aided positioning in gnss receivers: a performance analysis, *Signal Process.* 176 (2020) 107713.
- [27] F. Vincent, E. Chaumette, C. Charbonnieras, J. Israel, L. Ries, M. Aubourg, F. Barbiero, Asymptotically efficient gnss trilateration, *Signal Process.* 133 (2007) 270–277.
- [28] F. Vincent, O. Besson, Estimating time-varying DOA and doppler shift in radar array processing, *IEE Proc. Radar Sonar Navigat.* 147 (6) (2000) 285–290.
- [29] O. Besson, P. Stoica, Decoupled estimation of doa and angular spread for a spatially distributed source, *IEEE Trans. Signal Process.* 48 (7) (2000) 1872–1882, doi:10.1109/78.847774.
- [30] H.L. Van Trees, *Optimum Array Processing*, John Wiley, New York, 2002.
- [31] J. Liu, K. Li, X. Zhang, M. Liu, W. Liu, A weighted detector for mismatched subspace signals, *Signal Process.* 140 (2017) 110–115.
- [32] A. Coluccia, A. Fascista, G. Ricci, Cfar feature plane: a novel framework for the analysis and design of radar detectors, *IEEE Trans. Signal Process.* 68 (2020) 3903–3916.
- [33] N.B. Pulsone, C.M. Rader, Adaptive beamformer orthogonal rejection test, *IEEE Trans. Signal Process.* 49 (2001) 521–529.
- [34] E. Truslow, Performance Evaluation of the Adaptive Cosine Estimator Detector for Hyperspectral Imaging Applications, Northeastern University Boston, Massachusetts, 2012 Ph.D. thesis.
- [35] R.S. DiPietro, The Detection of Sub-pixel Objects and Mitigation of False Alarms in Hyperspectral Imagery, Northeastern University Boston, Massachusetts, 2010 Ph.D. thesis.
- [36] R.S. DiPietro, D. Manolakis, R.B. Lockwood, T. Cooley, J. Jacobson, Hyperspectral matched filter with false-alarm mitigation, *Opt. Eng.* 51 (1) (2012) 1–8.
- [37] J.W. Boardman, F.A. Kruse, Analysis of imaging spectrometer data using n-dimensional geometry and a mixture-tuned matched filtering approach, *IEEE Trans. Geosci. Remote Sens.* 49 (11) (2011) 4138–4152.
- [38] F. Vincent, O. Besson, Robust adaptive target detection in hyperspectral imaging, *Signal Process.* 181 (2021) 107905.
- [39] N. Acito, S. Matteoli, A. Rossi, M. Diani, G. Corsini, Hyperspectral airborne viareggio 2013 trial data collection for detection algorithm assessment, *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* 9 (6) (2016) 2356–2376.
- [40] M. Shi, G. Healey, Hyperspectral texture recognition using a multiscale opposite representation, *IEEE Trans. Geosci. Remote Sens.* 41 (5) (2003) 1090–1095.
- [41] E. Kelly, An adaptive detection algorithm, *IEEE Trans. Aerospace Electron. Syst.* 22 (2) (1986) 115–127.
- [42] O. Besson, F. Vincent, Sub-pixel detection in hyperspectral imaging with elliptically contoured t-distributed background, *Signal Process.* 175 (2020) 107662.
- [43] O. Besson, F. Vincent, S. Matteoli, Adaptive target detection in hyperspectral imaging from two sets of training samples with different means, *Signal Process.* 181 (2021) 107909.
- [44] E. Conte, M. Lops, G. Ricci, Asymptotically optimum radar detection in com-

- pound-gaussian clutter, *IEEE Trans. Aerospace Electron. Syst.* 31 (2) (1995) 617–625.
- [45] L.L. Scharf, T. McWhorter, Adaptive matched subspace detectors and adaptive coherence estimators, in: *Proceedings 30th Asilomar Conference Signals Systems Computers*, 1996, pp. 1114–1117. Pacific Grove, CA
- [46] F. Vincent, O. Besson, Non zero mean adaptive cosine estimator and application to hyperspectral imaging, *IEEE Signal Process. Lett.* 27 (2020) 1989–1993.
- [47] F. Vincent, O. Besson, Generalized likelihood ratio test for modified replacement model in hyperspectral imaging detection, *Signal Process.* 174 (2020) 107643.